CONFIDENTIAL • APRIL 2026

Card Network

ISO 8583 Payment Network — Capability Overview

Real-time authorization · Fraud detection · Interchange pricing · Clearing & Settlement · Native Agentic Commerce

4	8	7	75	080/081
Message Types	Fraud Rules	Fee Rate Rules	Tests Passing	Agent Sign-On/Off Agents register natively over ISO 8583 — no HTTP overlay

OVERVIEW

What Is This?

A fully functional, production-grade **ISO 8583 payment switch** — the same protocol used by Visa, Mastercard, and every major card network. It accepts TCP connections from acquirer banks, routes authorization requests to issuer banks, applies real-time fraud scoring, runs end-of-day clearing and settlement, calculates **interchange and network fees** per transaction, and natively supports **AI agent payments** embedded in the ISO 8583 wire protocol.

Real-time Switch

Netty TCP server on port 8583 handles concurrent ISO 8583 connections. Sub-millisecond routing for 200+ simultaneous transactions.

Fraud Engine

8 configurable fraud rules with score-based aggregation — including 3 agent-aware rules. Decisions in <1ms, before any issuer contact.

Agentic Commerce

AI agents sign on via ISO 8583 NMC 080/081. Spend controls, MCC scope, intent hash, and multi-agent chain auth are carried natively in DE48 — no HTTP overlay needed.

Interchange Pricing

Per-transaction interchange and network fees calculated at clearing. MCC-based rate lookup, configurable schedule, applied to net settlement.

Clearing & Settlement

Daily batch clearing at 23:00 UTC, net position settlement at 23:30 UTC. Net position includes interchange and network fee adjustments.

Operations Dashboard

Browser-based admin UI with live transaction monitoring, fraud alert review queue, fee schedule management, agent registry, and manual clearing controls.

AWS-Ready

Terraform IaC for ECS Fargate multi-AZ deployment. Aurora PostgreSQL, ALB + NLB, ShedLock for distributed jobs.

ISO 8583 Transaction Flow

AUTHORIZATION (0100 → 0110)

Acquirer Bank → **0100 Request** (PAN · Amount · MCC · DE48) → **Spend Controls** (DE48 Agent Check) → **Fraud Engine** (Score · Decide) → **BIN Router** (PAN → Issuer) → **0110 Response** (RC · Auth ID · DE48 echo) → Acquirer Bank

END-OF-DAY PROCESSING

Approved Txns → **ClearingEngine** (batch records) → **InterchangeService** (MCC → rate → fee) → **SettlementEngine** (net positions) → **SettlementRecords** (fee breakdown + intent_hash)

0100 / 0110
Authorization
Request & Response
implemented

0200 / 0210
Financial
Request & Response
implemented

0400 / 0420
Reversal
Request & Advice
implemented

0800 / 0810
Network Mgmt
Sign-on · Echo · Agent NMC
080/081
implemented

KEY ISO 8583 DATA ELEMENTS

- **DE2** — Primary Account Number (PAN)
- **DE3** — Processing Code
- **DE4** — Transaction Amount
- **DE18** — MCC → fee rate lookup
- **DE32/33** — Institution Codes
- **DE37** — Retrieval Reference Number
- **DE39** — Response Code
- **DE70** — NMC (Network Mgmt Code)
- **DE48** — Additional Data (Agent Context)

FRAUD DETECTION

Real-time Fraud Engine — 8 Rules

HOW IT WORKS

Every transaction is scored **before** it reaches the issuer. Each enabled rule contributes a weight to the aggregate score (0–100). Agent-aware rules additionally inspect DE48 context fields.

Score Range	Action	RC
≥ 80	DECLINE	59 — Suspected Fraud
≥ 50	FLAG	Continue to issuer + alert
< 50	APPROVE	Continue normally

5 CORE RULES + 3 AGENT-AWARE RULES

Rule	Trigger	Wt
Velocity	Max N txns per card per hour	25
Amount Limit	Amount above $5k threshold	30
Blacklist	SHA-256 PAN in blacklist → instant DECLINE	100
Decline Velocity	≥3 declines / hour → card-testing	60
MCC Risk	High-risk MCC (gambling, pawnshops, crypto)	20
Agent Velocity	>10 agent txns in 5 min window	35
Agent Scope	MCC not in DE48 SCP scope list	45
Agent Chain	CHN depth >3 or unknown sub-agent	40

FRAUD OPERATIONS

DB-Driven Config

Rules stored in PostgreSQL with JSON parameters. Enable/disable and adjust thresholds without redeployment. Cache refreshes every 60s.

Alert Queue

Every flagged or declined transaction creates a FraudAlert record. Dashboard shows a review queue with fraud score and triggered rule names.

PAN Blacklist

PANs added as SHA-256 hashes — raw card numbers are never stored. Supports optional expiry timestamps for temporary blocks.

CLEARING & SETTLEMENT

End-of-Day Processing

DAILY SCHEDULE

Time (UTC)	Process	Output
23:00	Clearing	ClearingBatch · ClearingRecords with fees per txn
23:30	Settlement	Net position per participant including fee adjustments

ShedLock (Multi-Instance)

PostgreSQL-backed distributed lock ensures only one ECS task executes clearing or settlement at a time across any number of instances.

SETTLEMENT NET POSITION FORMULA

```
credit = Σ amounts where participant = issuer
debit = Σ amounts where participant = acquirer

iReceived = Σ interchange received (as issuer)
iPaid = Σ interchange paid (as acquirer)
nFees = Σ network fees paid (both sides)

net = credit − debit + iReceived − iPaid − nFees
```

- Issuers gain interchange income from every transaction they issue
- Acquirers pay interchange on every transaction they process
- Both sides pay a scheme (network) fee per transaction
- One SettlementRecord per participant — stores full fee breakdown

SUPPORTED CONTROLS

Manual Trigger

Dashboard "Run Clearing" button and `POST /clearing/run` endpoint allow out-of-schedule runs for testing.

Idempotency

Clearing is idempotent per date. Re-running for a date with a COMPLETE batch is a safe no-op.

CSV Export

`GET /settlement/{date}/report` and `GET /api/interchange/report?format=csv` for full fee breakdown.

Interchange & Network Fee Engine

HOW FEES ARE CALCULATED

```
fee = round(amount × bps / 10,000) + flat

Example: $100.00, MCC 5411 (retail rate)
interchange = round(10000 × 120 / 10000) + 8 = 128 ¢ ($1.28)
network fee = round(10000 × 5 / 10000) + 2 = 7 ¢ ($0.07)

Example: $100.00, MCC 7995 (high-risk)
interchange = round(10000 × 250 / 10000) + 15 = 265 ¢ ($2.65)
```

Rate Matching Logic

1. Look up exact MCC in `interchange_rates` (highest priority wins)
2. Fall back to DEFAULT catch-all rate if no exact match
3. Return 0 if no rate configured — never blocks clearing

TWO FEE CATEGORIES

Category	Direction	Applied To
INTERCHANGE	Acquirer → Issuer	Every cleared transaction
NETWORK	Both → Network	Both sides pay per transaction

MCC Persistence

DE18 (Merchant Category Code) is captured from every inbound ISO 8583 message and stored on the Transaction record. At clearing time, InterchangeService uses the MCC for rate lookup.

DEFAULT RATE SCHEDULE (7 SEEDED RULES)

Name	Category	MCC Pattern	Rate (bps)	Flat	Priority	Description
Interchange Standard	INTERCHANGE	DEFAULT	150 bps (1.50%)	$0.10	0	Catch-all rate for unmatched MCCs
Interchange Retail	INTERCHANGE	5411	120 bps (1.20%)	$0.08	5	Grocery & retail — preferred rate
Interchange High-Risk	INTERCHANGE	7995	250 bps (2.50%)	$0.15	10	Gambling / betting
Interchange High-Risk	INTERCHANGE	5933	250 bps (2.50%)	$0.15	10	Pawnshops
Interchange High-Risk	INTERCHANGE	6051	250 bps (2.50%)	$0.15	10	Crypto & non-bank financial
Network Fee — Acquirer	NETWORK	DEFAULT	5 bps (0.05%)	$0.02	0	Scheme fee — acquirer side

MANAGEMENT INTERFACES

REST API & Admin Dashboard

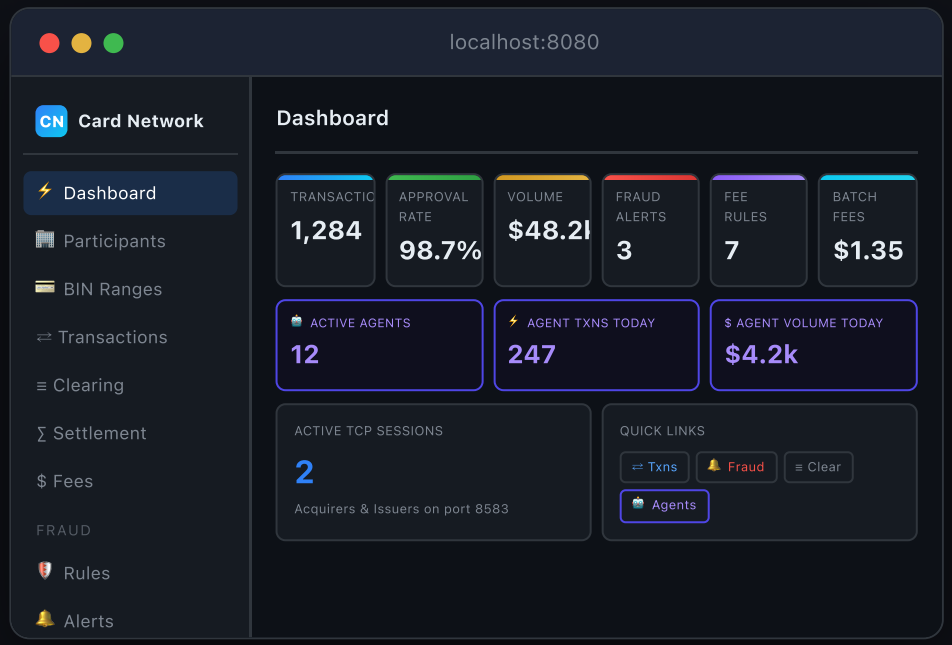
REST API (HTTP BASIC AUTH)

Domain	Operations
Participants	CRUD — create, read, update, deactivate
BIN Ranges	List, create, delete routing entries
Transactions	Paginated list, detail view
Fraud Rules	Full CRUD with live JSON parameter editing
Fraud Alerts	Paginated list, mark reviewed
Blacklist	Add (raw PAN → hash), list, remove
Clearing	List batches, trigger run
Settlement	List records, CSV export with fee columns
Interchange	CRUD rates · Fee report (JSON + CSV)
Agents	Register · list · update · stats (active, txns today, volume)
Test Lab	List scenarios, run, view results

Security Model

Dashboard: Spring Security form login · REST: HTTP Basic
CSRF disabled for `/api/**` · PANs masked (first 6 + last 4)

DASHBOARD UI (DARK THEME · BOOTSTRAP ICONS)



Dark theme (GitHub palette) · Bootstrap Icons nav · Gradient accent stat cards · Agent Commerce row · 14 pages

EDGECASESUITETEST — 12 SCENARIOS

Scenario	What It Validates
Happy Path	End-to-end approval, auth ID generated
Issuer Decline	RC=05 propagated correctly to acquirer
BIN Not Found	RC=15 when no BIN range matches
Blacklisted Card	Fraud DECLINE (RC=59), alert created
High-Risk MCC	MCC=7995 → fraud FLAG, issuer approves
Large Amount	Amount > $5k → fraud FLAG
Velocity Breach	7 rapid txns trigger velocity alert
Decline Velocity	4 declines → DECLINE (RC=59)
Echo Test	0800 NMC=301 → 0810 RC=00 immediately
Reversal	Approve then 0400 → 0420 RC=00, status REVERSED
Volume Load	**200 sequential transactions**, all approved
Mixed Batch	8 txns: happy + BIN miss + fraud + large amount

UNIT, INTEGRATION & AGENT TESTS

— **InterchangeRateTest** — calculateFee() math, rounding

— **InterchangeServiceTest** — MCC lookup, DEFAULT fallback, zero return

— **SettlementEngineTest** — Fee aggregation, net position formula

— **TransactionLoggerTest** — MCC captured from DE18

— **InterchangeControllerTest** — CRUD + CSV export

— **FeesIntegrationTest** — Flyway seeds 7 rates; end-to-end fee math

— **ClearingEngineTest** — Fees on records, totalFees on batch, MCC routing

— **MessageFactoryTest** — ISO 8583 encode/decode

— **BinRouterTest** — BIN range lookup edge cases

— **FraudDetectionServiceTest** — Rule scoring, thresholds

— **GatewayIntegrationTest** — Full TCP stack, Testcontainers PG

— **AuthorizationE2ETest** — Full auth flow: acquire → fraud → route → issue → respond

— **LiveAuthDemoTest** — Real-time 0100 approval with human-readable log output

— **AgentCommerceTest** — 5 ordered E2E tests: sign-on, approved auth (DE48 echoed), per-txn cap → RC=61, MCC scope → RC=57, sign-off → INACTIVE

— **VolumeTest** — 3 scenarios: 500 concurrent txns, 200-burst, 100 connection churn cycles

— **DurationTest** — 2 scenarios: 30-second sustained (≥60 txns), 5-client 20-second (≥100 txns)

Infrastructure

Testcontainers spins up real PostgreSQL with all Flyway migrations

PERFORMANCE

Volume & Duration Tests — 5 Tests, 0 Failures

VOLUMETEST — CONCURRENT LOAD

Scenario	Load	Pass Threshold
Concurrent 500	10 clients × 50 txns	≥ 490 approved (98%)
Burst 200	20 clients × 10 txns	≥ 196 approved (98%)
Connection Churn	100 sequential sign-on / off cycles	100 / 100 complete

DURATIONTEST — SUSTAINED THROUGHPUT

Scenario	Duration	Pass Threshold
Sustained 30 s	1 client, back-to-back, 30 seconds	≥ 60 txns · 0 failures
Multi-client 20 s	5 clients simultaneously, 20 seconds	≥ 100 txns (≥ 10 tps)

Race Condition — Found & Fixed

Writing concurrent tests immediately exposed a race in `ParticipantConnector.send()`: multiple threads each saw an empty session registry and called `bootstrap.connect().sync()` simultaneously. Only 1 connection was accepted by the fake issuer's `ServerSocket`; the remaining N−1 were zombie channels. Last `register()` overwrote the registry with a zombie → only 1 transaction approved, all others timed out.

Fix: Per-issuer double-checked locking with `ConcurrentHashMap<String, Object>` lock objects. The check-then-connect sequence is now atomic per issuer code. All 5 performance tests pass. Run time dropped from 17 min → 2:40 min.

Test Infrastructure

Each class runs a full Spring context with Testcontainers PostgreSQL. A raw Java `ServerSocket` plays the issuer, parsing ISO 8583 frames and replying with 0110 approvals. Participants sign on with 0800 NMC=001 before the test loop. Unique gateway ports (18590 / 18591) prevent inter-class interference.

Outcome	Condition	Dispute Status
ACCEPTED	Issuer responds RC=00	ACCEPTED
REJECTED_BY_ISSUER	Issuer responds RC≠00	REJECTED_BY_ISSUER
TIMEOUT	No 0210 within timeout window	RC=91 → PENDING

DATABASE — V11 FLYWAY MIGRATION

```
CREATE TABLE disputes (
  id BIGSERIAL PRIMARY KEY,
  transaction_id BIGINT REFERENCES transactions(id),
  claimant_id BIGINT REFERENCES participants(id),
  reason_code VARCHAR(10) NOT NULL,
  status VARCHAR(30) NOT NULL,
  amount BIGINT NOT NULL,
  description TEXT,
  created_at TIMESTAMPTZ DEFAULT now(),
  updated_at TIMESTAMPTZ DEFAULT now()
);
```

↓

ARBITRATION

↓

WON | LOST | WITHDRAWN

ADMIN DASHBOARD & REST API

Dashboard UI

- `GET /disputes` — list with status filter dropdown & pagination
- `GET /disputes/{id}` — lifecycle detail view with action buttons
- Stat card on main dashboard: **Open Disputes** count with purple accent

REST API (HTTP Basic)

- GET `/api/disputes` — paginated list, filter by status
- POST `/api/disputes` — file new chargeback
- PUT `/api/disputes/{id}` — advance lifecycle state
- GET `/api/disputes/stats` — count by status, total disputed amount

REASON CODE REFERENCE

VISA REASON CODES

Code	Category	Description
10.1	EMV Liability	Counterfeit — chip card used at mag-stripe

MASTERCARD REASON CODES

Code	Category	Description
4808	Authorization	Required authorization not obtained

making agentic commerce a first-class citizen of the payment protocol.

DE48 AGENT CONTEXT — WIRE FORMAT

```
AGT:AGENT001  ← Agent identifier (max 16 chars)
|INT:a3f9b1c2d4e5...  ← SHA-256 intent hash (64 hex)
|CHN:ORCHESTRATOR>AGENT001  ← Agent chain (>-delimited)
|SCP:5411,5812,4111  ← Allowed MCC list (comma-sep)
|LMT:10000  ← Per-transaction cap (cents)
|WIN:09:00-17:00  ← Allowed time window (HH:mm)
|NCE:7f3a9b2c  ← Nonce (replay prevention)
```

DE48 Field Properties

Packager type: `IFB_LLLCHAR` — variable length up to 999 chars. Carried transparently through the network. Echoed back verbatim in the 0110 authorization response. Persisted to the `transactions` table as three columns: `agent_id`, `intent_hash`, `agent_chain`.

NMC 080/081 — AGENT SIGN-ON / SIGN-OFF

Critical: DB transaction commits via `TransactionTemplate` *before*

AGENTREGISTRATION — PERSISTENT REGISTRY

Field	Type	Description
agent_id	VARCHAR(16)	Unique identifier, max 16 chars
participant_id	UUID FK	Owning acquirer participant
public_key	TEXT	Agent public key for audit
mcc_scope	VARCHAR(255)	Registration-level MCC allowlist
per_txn_limit	BIGINT	Per-transaction cap (cents)
daily_limit	BIGINT	Rolling 24h spend cap (cents)
time_window	VARCHAR(50)	Allowed hours e.g. 09:00-17:00
status	ENUM	ACTIVE / INACTIVE / SUSPENDED
last_seen_at	TIMESTAMPTZ	Updated on every approved txn
expires_at	TIMESTAMPTZ	Optional delegation expiry

SHA-256 INTENT HASH

The `INT` field carries a 64-character hex hash of the agent's stated purchase intent (e.g. *"buy 3 units of SKU-42 at $33.00 each"*). It is persisted to `transactions.intent_hash` and copied to settlement records at clearing time — creating an **immutable, cryptographically-bound audit trail** linking every monetary movement back to the agent's original intent. The network records the hash; the agent self-signs the payload off-chain.

MULTI-AGENT CHAIN AUTHENTICATION

Control	Source	Violation RC	Short-circuits?
Per-Transaction Cap	DE48 LMT field	RC=61 Exceeds Limit	Yes — immediate 0110
Daily Rolling Limit	AgentRegistration.dailyLimit	RC=61 Exceeds Limit	Yes — immediate 0110
MCC Scope	DE48 SCP / Registration	RC=57 Not Permitted	Yes — immediate 0110
Time Window	DE48 WIN field	RC=54 Delegation Expired	Yes — immediate 0110

MCC Scope Precedence

Registration-level scope (set at sign-on) **overrides** DE48 per-transaction scope when both are present — the registered allowlist is the authoritative source of truth. The DE48 scope is checked as a secondary constraint if the registration has no scope set.

Agent Scope Violation	AGENT_SCOPE	SCP field present, MCC mismatch	+45	FLAG
Agent Chain Anomaly	AGENT_CHAIN	Chain depth >3 or unknown agent	+40	FLAG

Rules are stored in `fraud_rule_configs` with JSON parameters and seeded by V12 migration. They are auto-discovered by `FraudDetectionService` via Spring component scanning — no code changes needed to add new rules.

REST API — AGENT MANAGEMENT

Method	Endpoint	Action
GET	/api/agents	Paginated list; filter by status
GET	/api/agents/{agentId}	Full agent detail + limits
POST	/api/agents	Register agent programmatically
PUT	/api/agents/{agentId}	Update status, limits, MCC scope
GET	/api/agents/stats	Active count, txns today, volume today

VS. AMEX AGENTIC COMMERCE EXPERIENCES (ACE)

Feature	Amex ACE	This Network
Protocol layer	HTTP overlay on card rails	Native ISO 8583 DE48
API access	Gated / invite-only	Open REST API
Multi-agent chain	Not supported	CHN field, depth enforcement

System Architecture

PARTICIPANTS

Acquirer Banks
TCP connect port 8583

Issuer Banks
TCP connect port 8583

AI Agents
NMC 080/081 sign-on, DE48 in every txn

TCP GATEWAY (NETTY)

GatewayServer
2-byte length-prefix framing · jPOS packager · SessionRegistry

GatewayMessageHandler
Routes by MTI → Auth / Financial / Reversal / NMC service

BUSINESS LOGIC

FraudDetectionService
8 rules · Score aggregation · DB-cached config

AgentRegistrationService
NMC 080/081 · DE48 spend controls · daily limit · MCC scope · time window

BinRouter
PAN → BinRange → Issuer

InterchangeService
MCC → rate lookup · INTERCHANGE + NETWORK fees per txn

ClearingEngine · SettlementEngine
Daily batch jobs · ShedLock · fee-adjusted net positions · intent_hash copied

PERSISTENCE

PostgreSQL 16
12 migrations · 12 entities · agent_registrations table

Spring Data JPA
12 repositories · Hikari pool

TECHNOLOGY STACK

Java 21 + Spring Boot 3.4
Virtual threads ready, modern Spring ecosystem

jPOS 2.1.9
ISO 8583 encode/decode via xml packager config; DE48 IFB_LLLCHAR for agent context

Netty 4.1
Non-blocking TCP gateway, async handlers

Flyway 11
12 versioned migrations; V12 adds agent_registrations + 3 agent fraud rule seeds

GAPS & ROADMAP

What's Still Missing

CORE PROTOCOL GAPS

- **Duplicate / Retransmission Detection**
  No idempotency on STAN. A retransmitted 0100 with the same STAN would create a duplicate transaction.

- **Batch File Support (ISO 8583 / ACH)**
  Only real-time TCP. No file-based clearing upload or bulk authorization advices.

- **3D Secure / PSD2 SCA**
  No authentication challenge flow (DE55 / EMV data, liability shift logic).

- **Tiered / Qualified Interchange**
  Real networks apply multi-tier rates (qualified / non-qualified / rewards). Current model uses a single rate per MCC with no card-type differentiation.

- **Agent Credential Verification**
  DE48 NCE nonce is recorded but not cryptographically verified. A full implementation would validate agent signatures off-chain before authorizing.

OPERATIONAL GAPS

- **Role-Based Access Control (RBAC)**
  Single admin credential. No read-only analyst role, no per-participant portal login, no API key management.

- **Audit Logging**
  Fraud rule changes, participant config edits, interchange rate changes, and blacklist additions are not persisted to an audit trail.

- **Real-time Reporting & Charts**
  Dashboard shows daily totals only. No hourly charts, trend lines, fraud rate over time, or agent commerce trend analysis.

- **Distributed Tracing (OTEL / Zipkin)**
  No trace IDs propagated through multi-hop flows. Debugging a slow issuer or tracing an agent chain requires log correlation by STAN.

- **Dead-Letter Queue & Retry**
  Issuer timeouts respond immediately with RC=91. No retry queue or delayed re-delivery for transient failures.

BUILT & WORKING

— Full ISO 8583 switch (0100, 0200, 0400, 0800)

— Real-time fraud engine — 8 configurable rules (5 core + 3 agent-aware)

— BIN routing — PAN → issuer bank

— Interchange & network fee calculation — 7 seeded rates

— Clearing & settlement with fee-adjusted net positions

— Fee schedule management & per-participant fee report

— Chargeback & dispute handling — full lifecycle, reason codes, REST API

— **Native agentic commerce** — DE48 agent context, NMC 080/081 sign-on/off

— Network-level spend controls — per-txn cap, daily limit, MCC scope, time window

— SHA-256 intent hash anchored in transactions & settlement records

— Multi-agent chain authentication (CHN field, depth enforcement)

— Agent-specific chargeback codes AGT-001 through AGT-004

— Operations dashboard — 14 pages including agent registry & agent commerce stats

— REST API — 40+ endpoints including /api/agents

— Interactive Test Lab — 12 scenarios

— 75 automated tests (unit + integration + agent + volume + duration)

— Terraform IaC for AWS ECS Fargate multi-AZ

NEXT PRIORITIES

— Duplicate / retransmission detection on STAN

— Agent credential cryptographic verification (nonce + signature)

— Tiered interchange (qualified / non-qualified / rewards)

— Role-based access control

— Audit trail for config changes

— Real-time charts & trend reporting (incl. agent commerce analytics)

— 3D Secure / EMV authentication data

— Distributed tracing (OTEL)

— HTTPS + ACM cert (AWS deployment)